

# Sai Pavan Kalyan Jampuram

+1-205-787-2477 | jampurams@gmail.com | [Linkedin](#) | [Github](#)

## EXECUTIVE SUMMARY

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Detail-oriented Data Analyst skilled in SQL, Python, Power BI, and predictive analytics. Experienced in building dashboards, optimizing data pipelines, and applying machine learning to solve real-world problems. Internship experience with EY and UAB in delivering measurable business outcomes. Strong grasp of data wrangling, automation, and visualization with a foundation in cloud technologies.

## TECHNICAL SKILLS

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- **Programming & Databases:** SQL (MySQL, PostgreSQL, SQL Server), Python (Pandas, NumPy, Scikit-learn), R
- **Data Visualization & BI Tools:** Power BI, Excel (Pivot Tables, Power Query), Matplotlib, Seaborn
- **Data Analytics & Business Intelligence:** KPI Dashboards, A/B Testing, ETL, Data Wrangling, Statistical Analysis
- **Machine Learning & AI (Basic):** Predictive Modeling, Regression, Classification, Anomaly Detection
- **Big Data & Cloud:** Microsoft Azure (Azure Data Factory, Synapse Analytics), AWS (S3, Lambda)
- **Other Tools:** Git, Agile Methodologies, Jira

## EXPERIENCE

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**Data Science Intern** | University of Alabama at Birmingham (UAB) – UAB Athletes Jan 2024 – May 2024

- Analyzed 500,000+ data points from GPS trackers, biometric sensors, and game stats to extract actionable insights.
- Developed a machine learning model (85% accuracy) to predict injury risks, reducing preventable injuries by 10%.
- Built 3 Power BI dashboards for real-time athlete performance tracking, improving training efficiency by 15%.
- Optimized SQL queries, reducing data retrieval time by 40%, enabling faster decision-making for coaches.
- Applied clustering algorithms to classify players based on fatigue levels, optimizing recovery plans.
- Automated data pipelines, cutting manual processing time by 30%, streamlining analytics workflows.
- Presented key findings to coaching staff and sports analysts, influencing data-driven game strategies.

**Data Science Intern** | Tiger Analytics | Chennai, India Jul 2022 - Oct 2022

- Automated 90% of data pipeline processes using SQL and Python, reducing processing time by 30%.
- Conducted exploratory data analysis (EDA) on 100,000+ records, identifying key business trends that increase marketing ROI by 20%.
- Built interactive Power BI dashboards, reducing report generation time by 25% and improving decision-making.
- Developed customer segmentation models (K-Means, DBSCAN) to predict behavior patterns, leading to a 15% increase in targeted marketing effectiveness.

**Research Assistant Intern** | Anna University | Chennai, India Jan 2021 - Jan 2022

- Conducted data collection and preprocessing for research on predictive modeling and AI applications.
- Developed Python-based scripts for text analysis, extracting insights from qualitative data.
- Published findings on AI-driven analytics and forecasting models. and assisted with data collection efforts, ensuring accurate and comprehensive data for analysis
- Performed qualitative analysis on collected data to identify trends or patterns, providing insights that informed project decisions

## PROJECTS

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### *AI-Driven Pulmonary Imaging Analysis*

Mentored by Dr. Sandeep Bodduluri | UAB CLAIR Lab - May 2024

- Developed a convolutional neural network using TensorFlow to analyze 2,000+ chest CT scans, improving COPD detection accuracy by 18%.
- Engineered an automated image classification pipeline in Python, reducing manual review time by 60%.
- Extracted quantitative lung function markers using radiomics, contributing to early diagnosis workflows.
- Integrated predictive outputs into Power BI dashboards, enabling real-time insights for clinical teams.
- Collaborated with radiologists and AI specialists to validate models, achieving 90% sensitivity in identifying emphysema regions.

## ***Gene Network Modeling in Sysyem Biology***

Mentor: Dr. Shahid M. Mukhtar | UAB Department of Biology

-Dec 2023

- Constructed network models in Cytoscape to map 3,500+ gene interactions under environmental stress conditions.
- Analyzed gene expression datasets using R and Python, identifying 25 key regulatory nodes in transcription networks.
- Applied clustering algorithms to classify gene modules, revealing high-confidence pathways shared with human immunity.
- Reduced data preprocessing time by 40% through custom-built data wrangling scripts.
- Presented findings in lab seminars, contributing insights for potential translational applications in biomedical modeling.

## ***Tumor Detection using Neural Networks***

- Developed a binary classification model using Python to classify tumors as benign or malignant based on the University of Wisconsin Hospitals' breast cancer dataset.
- The aim was to build a binary classification model that can determine, from a variety of features, whether a tumor is benign (0) or malignant (1).
- Achieved 92% accuracy through the implementation of a Python-based neural network.

## ***Cardiovascular Risk Predictionusing EHR Data***

Mentor: Dr. April Carson | UAB School of Public Health

-Aug 2023

- Analyzed de-identified EHR data from 1,200+ patients to assess risk factors for hypertension and cardiovascular disease.
- Built a logistic regression model using R that predicted early cardiac risk with 87% accuracy.
- Extracted and cleaned 10,000+ clinical data points using SQL, streamlining model training datasets.
- Designed Power BI dashboards to visualize demographic disparities in heart disease, used in research briefings.
- Delivered insights that supported an NIH grant proposal on health disparities in underserved populations.

## ***The creation of a virtual machine on Microsoft Azure***

- Developed, and a NodeJS program was put into use to implement a REST API that uses the Express server to manage requests with GET, POST, PUT, and DELETE endpoints.
- Created an SQL database on Azure with a table named products and made sure that the virtual machine was the only way to access the database.
- The API was tested by sending requests to the virtual machine's public IP using Postman on a PC.
- Verified answers to CRUD queries on the database.

## **CERTIFICATIONS**

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- Google Data Analytics Professional Certificate
- IBM Data Science Professional Certificate
- Microsoft Power BI Certification
- AWS Cloud Practitioner Certification
- Certified Analytics Professional (CAP) – INFORMS

## **EDUCATION**

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### **University of Alabama at Birmingham - Birmingham, AL**

Master of Science, Electrical and Computer Engineering

May 2025

### **Anna University - Chennai. IN**

Bachelor of Science, Electrical, Electronics and Communications Engineering

Jul 2022